Particle Physics Note

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Chapter 1

Decay Rate & Cross Sections

§ 1.1 Fermi Golden Rule

Consider time-dependent Schrödinger equation,

$$i\frac{d\psi}{dt} = \hat{H}(t)\psi. \tag{1.1}$$

We can separate Hamiltonian operator into two parts, one is time-independent, and else is time-dependent,

$$\hat{H}(t) = \hat{H}_0 + \hat{H}'(\mathrm{x}, t) \tag{1.2}$$

For time-independent part we can use common method to solve it,

$$\hat{H}_0\psi_k = E_k\psi_k, \quad \text{and} \quad \langle\psi_j|\psi_k\rangle = \delta_{jk} \tag{1.3}$$

Then the wavefunction can be expressed in terms of complete set of states of the unperturbed Hamiltonian as

$$\psi(\mathrm{x}, t) = \sum_k c_k(t)\psi(\mathrm{x})e^{-iE_k t}. \tag{1.4}$$

Then we can substitute Eq. 1.4 into Eq. 1.1, we can get

$$\begin{aligned} i\sum_k \left[\frac{\mathrm{d}c_k}{\mathrm{d}t}\psi_k e^{-iE_k t} - iE_k c_k \psi_k e^{-iE_k t}\right] &= \sum_k c_k \hat{H}_0 \psi_k e^{-iE_k t} + \sum_k \hat{H}' c_k \psi_k e^{-iE_k t} \\ \Rightarrow \quad i\sum_k \frac{\mathrm{d}c_k}{\mathrm{d}t}\psi_k e^{-iE_k t} &= \sum_k \hat{H}' c_k(t)\psi_k e^{-iE_k t} \end{aligned} \tag{1.5}$$

Suppose the initial state is i for $\psi_i \equiv |i\rangle$, that is

$$\psi(\mathrm{x}, 0) = \sum_k c_k(0)\psi_k = c_i(0)\psi_i. \tag{1.6}$$

We can easily imply that $c_k = \delta_{ik}$. Assume that time-dependent Hamiltonian is perturbed, which implies $c_{i\neq k} \ll 1, c_i \sim 1$. And Eq. 1.5 can reduce to

$$i\sum_k \frac{\mathrm{d}c_k}{\mathrm{d}t} |k\rangle e^{-iE_k t} \approx \hat{H}' |i\rangle e^{-iE_i t} \tag{1.7}$$

Then we need to get to final state, so we can use relation of inner product, let $\langle f|$ act on Eq. 1.7, we can get the coefficient c_f

$$\frac{dc_f}{dt} = -i \langle f|\hat{H}'|i\rangle e^{-i(E_i-E_f)t}, \tag{1.8}$$

where $\langle f|\hat{H}'|i\rangle \equiv T_{fi}$ is **transition matrix element**, can be calculated by

$$\langle f|\hat{H}'|i\rangle = \int_V \psi_f^*(\mathrm{x})\hat{H}'\psi_i(\mathrm{x})\,\mathrm{d}^3\mathrm{x}\,. \tag{1.9}$$

Then we can calculate the coefficient c_f, at time $t = T$

$$c_f(T) = -i\int_0^T T_{fi}e^{-i(E_i-E_f)t}\,\mathrm{d}t \tag{1.10}$$

The probability for a transition to the state $|f\rangle$ is given by

$$P_{if} = c_f^*(t)c_f(t) = |T_{fi}|^2 \int_0^T\int_0^T e^{-i(E_i-E_f)t}e^{-i(E_i-E_f)t'}\,\mathrm{d}t\,\mathrm{d}t' \tag{1.11}$$

And we have **transition rate** $\mathrm{d}\Gamma_{fi}$ from the initial state $|i\rangle$ to the single final state $|f\rangle$

$$\mathrm{d}\Gamma_{fi} = \frac{P_{fi}}{T} = \frac{1}{T}|T_{fi}|^2 \int_{-\frac{T}{2}}^{\frac{T}{2}}\int_{-\frac{T}{2}}^{\frac{T}{2}} e^{-i(E_i-E_f)t}e^{-i(E_i-E_f)t'}\,\mathrm{d}t\,\mathrm{d}t' \tag{1.12}$$

If we consider the time is long enough comparing to transition process, we can take a limit,

$$\mathrm{d}\Gamma_{fi} = |T_{fi}|^2 \lim_{T\to\infty}\left\{\frac{1}{T}\int_{-\frac{T}{2}}^{+\frac{T}{2}}\int_{-\frac{T}{2}}^{+\frac{T}{2}} e^{i\left(E_f-E_i\right)t}e^{-i\left(E_f-E_i\right)t'}\mathrm{d}t\,\mathrm{d}t'\right\}. \tag{1.13}$$

Associated with the definition of Dirac delta-function, the integral over $\mathrm{d}t'$ can be replaced by $2\pi\delta(E_f - E_i)$ and thus

$$\Gamma_{fi} = 2\pi\,|T_{fi}|^2 \left|\frac{\mathrm{d}n}{\mathrm{d}E_f}\right|_{E_i}, \tag{1.14}$$

where $\left|\frac{\mathrm{d}n}{\mathrm{d}E_f}\right|_{E_i}$ is called **density of states**, it can be also written as

$$\rho(E_i) = \left|\frac{\mathrm{d}n}{\mathrm{d}E_f}\right|_{E_i} \tag{1.15}$$

Fermi's golden rule for the total transition rate is therefore

$$\Gamma_{fi} = 2\pi \left|T_{fi}\right|^2 \rho(E_i) \tag{1.16}$$

However, we take assumption $c_{i\neq f}(t) \approx 0$, if we need to get more precise information, we should expand more terms of transition matrix element, we start from Eq. 1.8

$$\frac{\mathrm{d}c_f}{\mathrm{d}t} \approx -i\langle f|\hat{H}|i\rangle e^{i\left(E_f-E_i\right)t} + (-i)^2 \sum_{k\neq i} \left\langle f \left|\hat{H}'\right| k\right\rangle e^{i\left(E_f-E_k\right)t} \int_0^t \left\langle k \left|\hat{H}'\right| i\right\rangle e^{i(E_k-E_i)t'} \mathrm{d}t' \tag{1.17}$$

Therefore, the improved approximation for the evolution of the coefficients $c_f(t)$ is given by

$$\frac{\mathrm{d}c_f}{\mathrm{d}t} = -i\left(\langle f|\hat{H}|i\rangle + \sum_{k\neq i} \frac{\left\langle f \left|\hat{H}'\right| k\right\rangle \left\langle k \left|\hat{H}'\right| i\right\rangle}{E_i - E_k}\right) e^{i\left(E_f-E_i\right)t}. \tag{1.18}$$

For second order of transition matrix element, we have

$$T_{fi} = \langle f|\hat{H}|i\rangle + \sum_{k\neq i} \frac{\left\langle f \left|\hat{H}'\right| k\right\rangle \left\langle k \left|\hat{H}'\right| i\right\rangle}{E_i - E_k} \tag{1.19}$$

§ 1.2 Decay Rate

Fermi's golden rule can be written as an alternative forma

$$\Gamma_{fi} = 2\pi \int \left|T_{fi}\right|^2 \delta(E_i - E)\, \mathrm{d}n \tag{1.20}$$

Firstly consider the decay rate for the process $a \to 1 + 2$ in non-relativistic situation, related Fermi's golden rule, we write transition matrix element

$$T_{fi} = \langle \psi_1\psi_2|\hat{H}'|\psi_a\rangle \tag{1.21}$$

$$= \int_V \psi_1^*\psi_2^*\hat{H}'\psi_a\, \mathrm{d}^3x \tag{1.22}$$

In the Born approximation, the perturbation is taken to be small and the initial- and final-state particles are represented by plane waves of the form

$$\psi(\mathrm{x}, t) = Ae^{i(\mathbf{p}\cdot)} \tag{1.23}$$

§ 1.3 Cross Sections

Chapter 2

Klein-Gordon Equation